

decarbonised ground (this work):
nuclear $\sim 22 \rightarrow$ solar+storage $\sim 91 \text{ g kWh}^{-1}$

This work · low mass
(10 yr, dawn–dusk, +non- CO_2 + CH_4)

31

This work · mid mass

41

This work · high mass

63

~ 20

Aili et al. 2025: orbital matches
all-renewable terrestrial grid (CUE metric)

7

Ohs et al. 2025 · solar only
(no eclipse battery)

52

Ohs et al. 2025 · Starship
(5 yr, +battery +re-entry)

66

Ohs et al. 2025 · Falcon-9

Gas-fired ground = 590 g kWh^{-1} (off scale)

0

20

40

60

80

100

120

Lifecycle GHG intensity ($\text{g CO}_2\text{e per kWh delivered}$)